

Smart Jaynamaz

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Abstract—Praying Salah is an essential part in our (Muslim's) life. Salah is a main part of Muslim rituals. So, we are building a Smart jaynamaz to count prostration so that we can check if there is any fault in salah. We also have tasbih system so that people can count dhikr. We also have more exciting features.

I. PROJECT OVERVIEW

The Smart Jaynamaz is an innovative electronic prayer mat designed to enhance the spiritual experience of Muslims during their prayers (Salah). By integrating various electronic components and sensors, this project aims to assist users in maintaining their prayer discipline while providing valuable feedback and features that promote mindfulness and safety.

Objectives:

- **Sijdah Counting:** Utilize Force Sensitive Resistors (FSR) to accurately count the number of Sijdahs performed during prayer. This feature allows users to track their prayer performance and recognize any deviations from the intended number.
- **Qibla Finder:** Incorporate a compass to guide users in finding the Qibla direction, ensuring that their prayers are aligned properly. This feature is essential for Muslims praying in unfamiliar locations.
- **Automated Folding Mechanism:** Employ servo motors to automatically fold the Jaynamaz after use, promoting convenience and ease of storage.
- **Tasbih Counter:** Integrate a digital tasbih counter to help users keep track of their Dhikr (remembrance of Allah) sessions, allowing them to maintain a focused spiritual practice.
- **Emergency Alert System:** Include a buzzer and an emergency button that users can press in case of medical emergencies, such as a heart attack. This feature will ensure prompt alerts for help, providing peace of mind to users.

II. COMPONENT LIST

The following components were used in this project:

- Arduino Uno X 1
- Force Sensitive Resistor X 2
- LCD display X 1
- 8 segmented display 4 digits X 1
- Rubber button X 1
- Servo motor X 4

A. Arduino Uno

The Arduino Uno is a popular microcontroller board designed for hobbyists, students, and professionals in electronics and programming. It is based on the ATmega328P microcontroller and provides a simple, user-friendly platform for developing embedded systems. The Uno is part of the larger Arduino family, and it is commonly used in projects like robotics, IoT devices, and automation.

Key Features:

- 1 Microcontroller: ATmega328P
- 2 Operating Voltage: 5V
- 3 Input Voltage (recommended): 7-12V
- 4 Digital I/O Pins: 14 (of which 6 can be PWM outputs)
- 5 Analog Input Pins: 6
- 6 Clock Speed: 16 MHz
- 7 Flash Memory: 32 KB (0.5 KB used by the bootloader)
- 8 SRAM: 2 KB
- 9 EEPROM: 1 KB
- 10 USB Connection: For programming and serial communication
- 11 Power Jack: For external power input
- 12 Reset Button: For restarting the program loaded onto the micro controller
- 13 Size: Compact and lightweight, suitable for mounting into various projects.

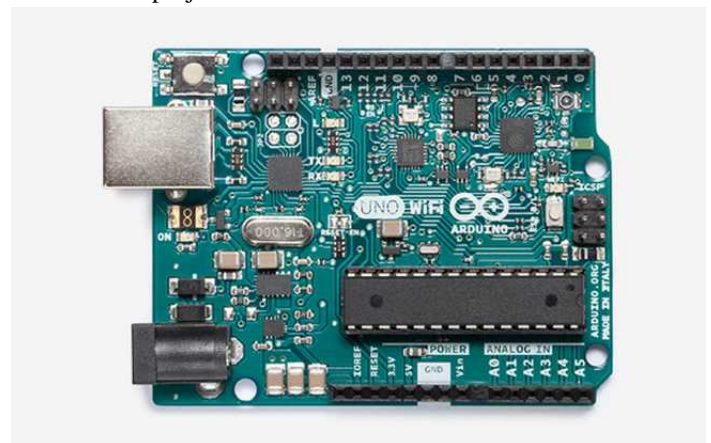


Fig. 1. Arduino uno

B. Force Sensitive Resistor (FSR)

A Force Sensitive Resistor (FSR) is a type of sensor that changes its resistance based on the amount of force or pressure applied to it. FSRs are widely used in various applications

where detecting touch, pressure, or weight is essential, such as robotics, medical devices, wearables, and interactive controls. The FSR consists of two layers: a substrate with conductive traces and a force-sensitive conductive polymer. When pressure is applied, the resistance between these layers' decreases, allowing for the measurement of the applied force. While FSRs are not highly precise, they are cost-effective, durable, and responsive, making them an excellent choice for a wide range of non-critical force-sensing applications.

Purpose: To count each prostration, we are using Force Sensitive Resistor. When the person bows down his/her hands will create a pressure on the FSR. thus, it will count one prostration using Arduino uno.

Features of Force Sensitive Resistor (FSR):

- 1) Pressure Sensitivity: FSRs vary their resistance based on the force applied. Higher pressure leads to lower resistance.
- 2) Thin and Flexible: These sensors are lightweight, thin, and flexible, allowing easy integration into compact or curved surfaces.
- 3) Wide Force Range: FSRs can measure a broad range of forces, from light touches (grams) to significant pressures (kilograms).
- 4) Durable: Designed to handle repeated force cycles without significant degradation, ensuring reliability over time.
- 5) Cost-effective: FSRs are affordable compared to other force-sensing technologies, making them accessible for various projects.
- 6) Rapid Response: FSRs respond quickly to changes in applied force, enabling real-time force detection in dynamic applications.
- 7) Low Power Consumption: They require minimal power, making them suitable for battery-powered and portable devices.
- 8) Non-linear Output: The sensor's output is non-linear, meaning the sensitivity changes depending on the applied force. This is more pronounced with light touches.
- 9) Easy to Interface: FSRs are straightforward to integrate into circuits, often using a simple voltage divider configuration connected to micro controllers or analog-to-digital converters.

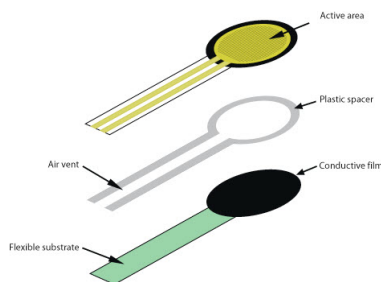


Fig. 2. Force Sensitive Resistor

C. 8-Segment Display

An 8-segment display is a type of electronic display device used to show numerical digits and some letters. It consists of

eight individual segments that can be illuminated in various combinations to represent different characters. These displays are commonly used in devices such as digital clocks, calculators, and various electronic instruments.

Purpose: To display the Tasbeih counting we are using this display.

Key Features of an 8-Segment Display:

- 1) Segment Configuration: The display typically consists of 7 segments arranged in a figure-eight shape, with an additional segment (the decimal point) for indicating decimal values. The segments are usually labeled from 'a' to 'g', and the decimal point is often labeled as 'dp'.
- 2) Character Representation: The combination of segments can represent numbers (0-9) and some letters (A-F), making them suitable for displaying hexadecimal values.
- 3) Common Cathode/Anode Types: Common Cathode: All the cathodes of the segments are connected to ground. To illuminate a segment, a positive voltage is applied to its anode. Common Anode: All the anodes are connected to a positive voltage. To illuminate a segment, ground is applied to its cathode.
- 4) Low Power Consumption: 8-segment displays are generally energy-efficient, consuming less power compared to other display technologies, which makes them suitable for battery-operated devices.
- 5) Simple Control: These displays can be easily controlled using simple logic circuits or microcontrollers, making them popular in DIY electronics and embedded systems.
- 6) Visibility: 8-segment displays are designed for high visibility and can be easily read from a distance, making them ideal for applications in dimly lit environments.
- 7) Multi-digit Configuration: Multiple 8-segment displays can be connected together to show larger numbers or multiple values (e.g., in digital clocks or timers)..
- 8) Variety of Colors: The segments can be made in various colors (e.g., red, green, blue) depending on the type of LED used, providing flexibility for aesthetic design in applications.



Fig. 3. 8-Segment Display

D. LCD Display

An LCD (Liquid Crystal Display) is a flat-panel display technology commonly used in devices such as televisions, computer monitors, smartphones, and many embedded

systems. LCD's use liquid crystals that modulate light in conjunction with a back light or reflector to produce images. Unlike older CRT displays, LCD's are energy-efficient, thin, lightweight, and can display images in high resolutions, making them ideal for modern electronic devices.

Purpose: To display all the information, we are using LCD display. like, showing the counting of prostration.

Key Features of LCD Display:

- 1) Thin and Lightweight: LCDs are significantly thinner and lighter than traditional CRTs, making them suitable for portable devices such as laptops, smartphones, and embedded systems.
- 2) Energy Efficient: LCDs consume less power compared to other display technologies like plasma, especially in devices with LED backlighting. This makes them ideal for battery-operated devices.
- 3) High Resolution: LCDs can display high-definition content with crisp, clear images, often supporting HD, Full HD, 4K, or higher resolutions.
- 4) Wide Range of Sizes: LCDs are available in various sizes, from small (1-2 inches) for embedded systems to large (up to 100+ inches) for televisions or billboards.
- 5) Backlit Technology: Most modern LCDs use LED backlighting to enhance brightness and contrast while improving energy efficiency. Earlier models used CCFL (Cold Cathode Fluorescent Lamps) for backlighting.
- 6) Color Display: LCDs can produce millions of colors by using liquid crystal cells that either block or allow light to pass through color filters, making them suitable for a wide range of applications requiring detailed color representation.
- 7) Touchscreen Capability: Many LCDs are available with touchscreen functionality, making them useful for smartphones, tablets, kiosks, and ATMs.
- 8) Wide Viewing Angle: Advanced LCDs (such as those using IPS – In-Plane Switching technology) provide wide viewing angles, ensuring that colors and image quality remain consistent even when viewed from the side.
- 9) Low Response Time: Modern LCDs offer fast response times, minimizing motion blur in videos or gaming applications, particularly for displays with higher refresh rates.
- 10) Versatility: LCDs can be found in a wide range of devices, including televisions, computer monitors, calculators, digital watches, and industrial equipment.

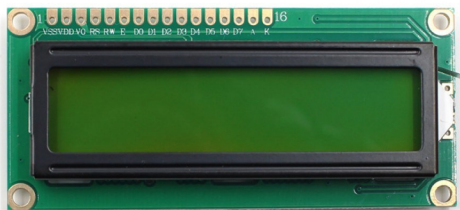


Fig. 4. LCD Display

E. Servo Motors

A servo motor is a type of motor that provides precise control of angular position, speed, and acceleration. Typically used in robotics, CNC machinery, and automated systems, servo motors are designed for high performance and efficiency.

Purpose: To fold the prayer mat, we are using 4 servo motors. each motors are connected with one wire which lifts a part of the prayer mat by 180 degrees. In this way we are folding the prayer mat.

Key Features of Servo Motor:

- 1) Precision Control: Ensures accurate positioning and movement.
- 2) Closed-Loop System: Continuously monitors and adjusts for minimal error.
- 3) High Torque-to-Weight Ratio: Efficient power in a compact size.
- 4) Speed Control: Offers variable and consistent speed settings.
- 5) Compact Size: Space-efficient design without sacrificing performance.
- 6) Types: Includes AC and DC options for various applications.
- 7) Feedback Mechanisms: Uses sensors for real-time position data.
- 8) Programmability: Can be programmed for specific tasks.
- 9) Durability: Designed for long-lasting and reliable operation.
- 10) Versatility: Suitable for a wide range of industries and applications.



Fig. 5. Servo Motor

F. Push Buttons

Push buttons are tactile switches made from flexible plastic material, commonly used in various electronic devices and

appliances. They provide a soft and durable interface for user interaction, allowing users to operate devices with a simple press. Push buttons are often found in remote controls, keyboards, medical devices, and consumer electronics due to their ease of use and reliability.

Purpose: To count Tasbih we are using soft button like this

Key Features of Push Button:

- 1) Actuation Mechanism: Mechanical or tactile feedback when pressed.
- 2) Durability: Designed for multiple presses, often rated for high cycle counts.
- 3) Size and Shape: Available in various sizes and shapes (round, square) to fit different applications.
- 4) Speed Control: Offers variable and consistent speed settings.
- 5) Material: Typically made from plastic, metal, or rubber for resilience and functionality.
- 6) Feedback Mechanisms: Uses sensors for real-time position data.
- 7) Programmability: Can be programmed for specific tasks.
- 8) Tactile Feel: Soft, clicky, or smooth press, offering different user experiences.



Fig. 6. Push Button

III. FEATURES

- 1) Qibla finding system.
- 2) Jaynamaz folding system.
- 3) Sijdah counting system.
- 4) Tasbeeh counting system.

Qibla finding system: The process starts when the user presses the power button, which powers up the Arduino. Then, four servos are activated to unfold the mat, and the LCD display is turned on. The LCD remains on for 20 seconds and displays the Qibla direction. After waiting for 20 seconds, the LCD turns off, and the system enters prayer mode, indicating the mat is ready for use. This process automates the preparation for prayer by guiding the user through the setup.

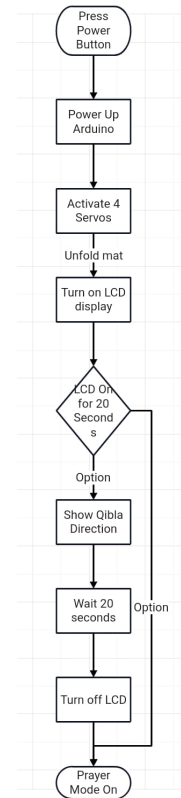


Fig. 7. Flow Chart for Qibla finding system

Sijdah counting system: The process starts when the user begins praying. After the prayer is completed, the user presses a Finish Button to signify the end of the prayer session. This action turns on a display, which then shows the total number of Sijdahs performed during the prayer. The flowchart outlines a simple, step-by-step process for recording and viewing prayer-related metrics.

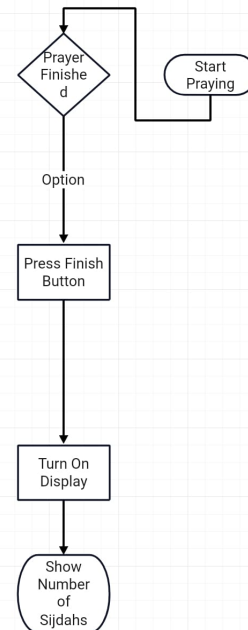


Fig. 8. Flow Chart for Sijdah counting system

Tasbeeh counting system: It begins when the user presses the Tasbih Button, which activates the device and displays the current count on the LCD screen. From there, the user has the option to press the Counting Button, which increases the Tasbih count with each press. The counter continues to increment as the user repeatedly presses the counting button, looping back to display the updated count. Additionally, the device features a Reset Button that, when pressed, resets the count and returns the device to its starting state. This straightforward process ensures that users can easily track and reset their Tasbih recitations.

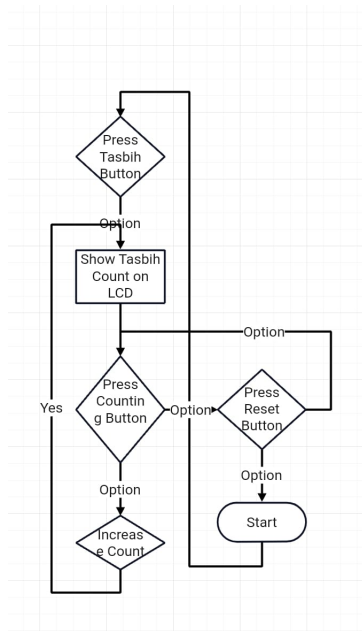


Fig. 9. Flow Chart for Tasbeeh counting system

IV. IMPLEMENTATION

Our project is implemented using C++ with the Arduino Programming Language, which is based on the Processing language and works with a modified version of the Processing IDE.